Alternative Grading in Introductory Computer Science Courses

Practices, Outcomes, and Challenges

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Tuesday 18 August 2026

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Outline

Introduction

Origin Story

- Tension

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- Tension
  - Opportunities to learn

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  - Ensuring understanding

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Origin Story

- Tension
  - Opportunities to learn
  - Ensuring understanding
- Allowed students to recover points from exams
- Changing relative weighting
- Geometric mean
- Challenge:
  - Recover from early mistakes
  - Grades reflect real understanding

Initial Approach

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- Specifications Grading

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- Projects due each week

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- Four levels for projects:

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  - I: Incomplete

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  - S: Satisfactory
  - P: Partial understanding
  - I: Incomplete
- Oops tokens

CSC 111 Fall 2018

CSC 111 Fall 2018 Grading Scheme

	A	B	C	D	F
Class meetings (28 total)	26	24	22	20	-
Classroom activities (28)	26	24	22	20	-
Chapter reading quizzes	90%	80%	70%	60%	-
Projects: S or E (14 total)	12	11	10	9	-
Projects: E	9	7	5	-	-

CSC 111 Fall 2020

Simplified Approach

- Two levels for projects: Pass or Not

CSC 111 Fall 2020

Simplified Approach

- Two levels for projects: Pass or Not
- Submit any two projects each week

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CSC 111 Fall 2020 Grading Scheme

	A	B	C	D	F
Projects (14 total)	13	12	10	9	-

CSC 101 Fall 2023 – Standards-Based Testing

Standards-Based Testing

- Classes focused on learning targets

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CSC 101 Fall 2023 Grading Scheme

	A	B	C	D	F
Learning Targets (out of 33 total)	30	27	24	20	-

Mixed Approach

CSC 101 Spring 2026 Grading Scheme

	A	A-	B+	B	B-	C+	C	C-	D	F
Learning Targets (32)	30	29	28	27	26	25	23	22	19	-
Online Homework							75%		60%	-
Computational Life	4			3			2		1	-
Resource Utilization									1	-

Mixed Approach

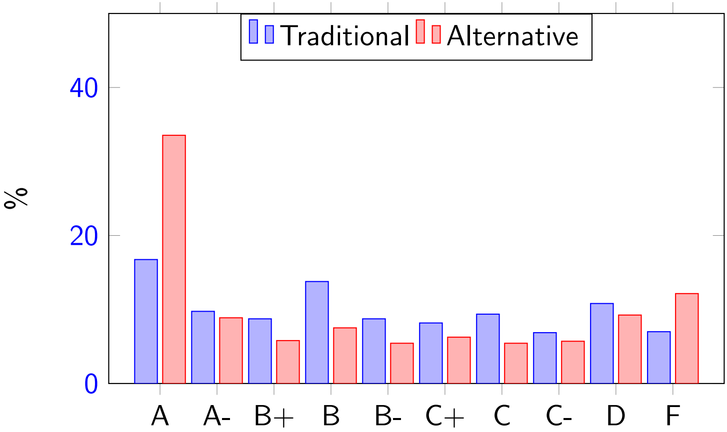
CSC 101 Spring 2026 Grading Scheme

	A	A-	B+	B	B-	C+	C	C-	D	F
Learning Targets (32)	30	29	28	27	26	25	23	22	19	-
Online Homework							75%		60%	-
Computational Life	4			3			2		1	-
Resource Utilization									1	-

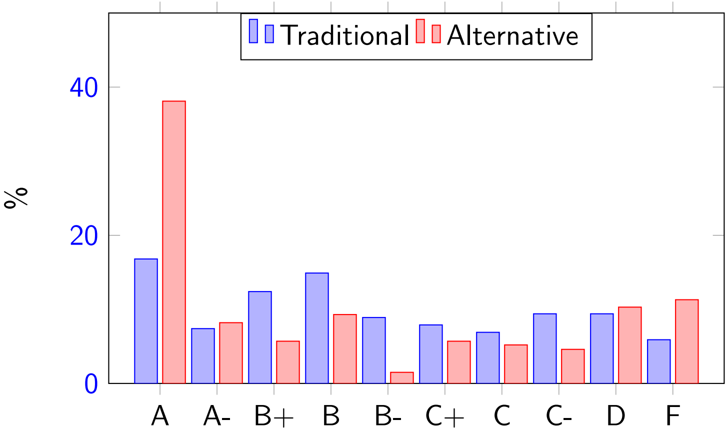
CSC 111 Spring 2026 Grading Scheme

	A	A-	B+	B	B-	C+	C	C-	D	F
Learning Targets (20)	19	18	17	16	15	14	13	12	11	-
Projects (13)	12		11		10		9	8	7	-
Online Homework							75%		60%	-
Computational Life	4			3			2		1	-
Resource Utilization									1	-

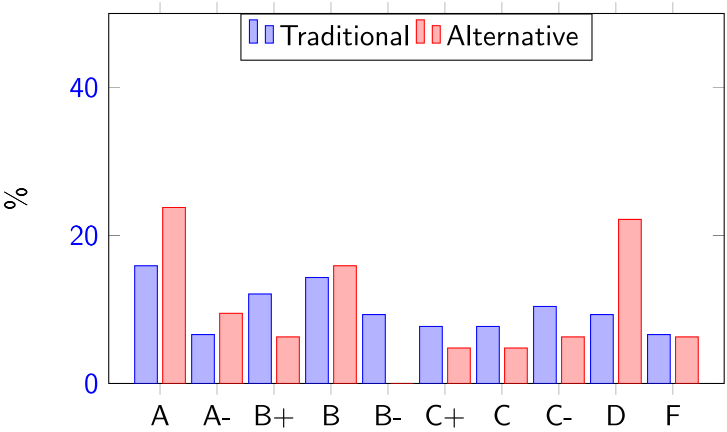
All Courses Grade Distribution



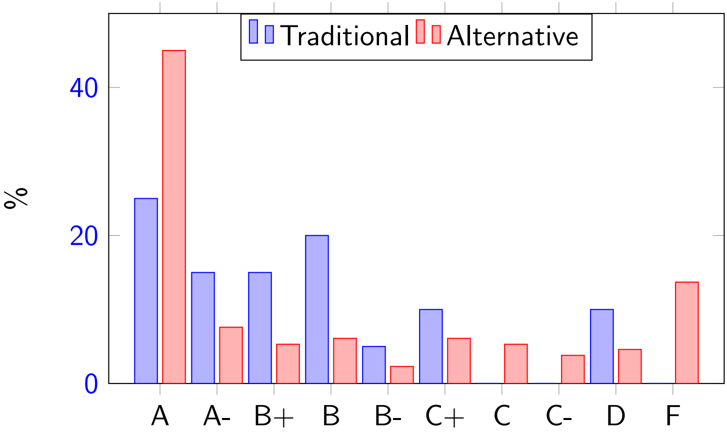
CSC 101 & 111 Grade Distribution



CSC 101 Grade Distribution



CSC 111 Grade Distribution



AI

AI resistant?

- Grades not averaged together

AI

AI resistant?

- Grades not averaged together
- Performance on projects cannot balance out low exam scores

AI

AI resistant?

- Grades not averaged together
- Performance on projects cannot balance out low exam scores
- Must still perform well on in-class assessments

Disadvantages

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- Procrastination

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- Procrastination
- Declining attendance in recent years

Possible Modifications

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- Required final exam

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- Attendance policy

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- Required final exam
- Attendance policy
- In-class assignments

Summary

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My experience suggests that alternative grading can provide a more accurate picture of student learning in introductory computer science courses, particularly when course grades depend on demonstrated mastery rather than accumulated points. At the same time, these systems require careful design to address student procrastination, attendance, and logistical complexity. Continued refinement of these approaches may help preserve their benefits while reducing their drawbacks.